

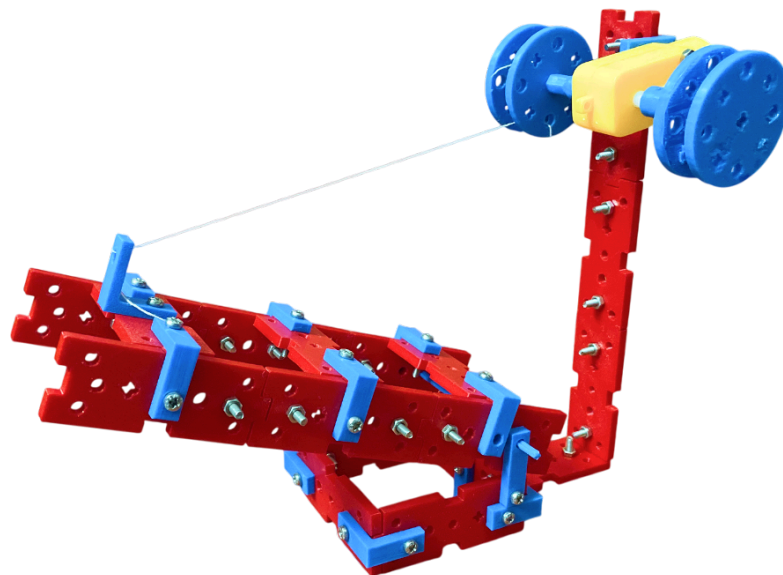
level 1

draw bridge

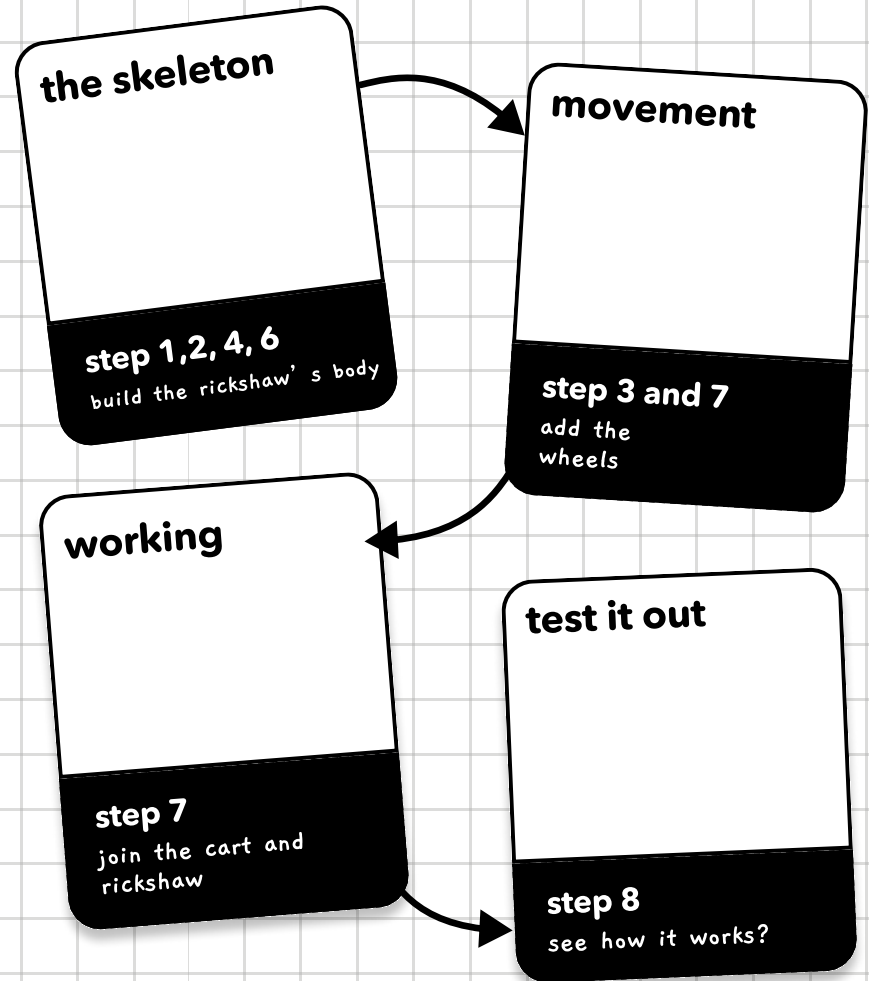
a clever machine that lifts to let boats pass! I use a pulley and axle system to raise and lower the bridge. When you turn the handle, my pulley pulls the bridge up, showing how simple machines work together to move big things with little effort!

no. of sessions: 6
simple machine:
axle & pulley

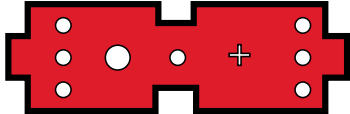
oh.



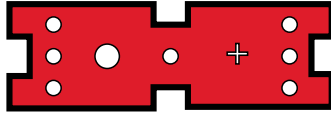
what's the goal?



materials needed



m-m beam - x4



f-f beam - x2



l clamp - x4

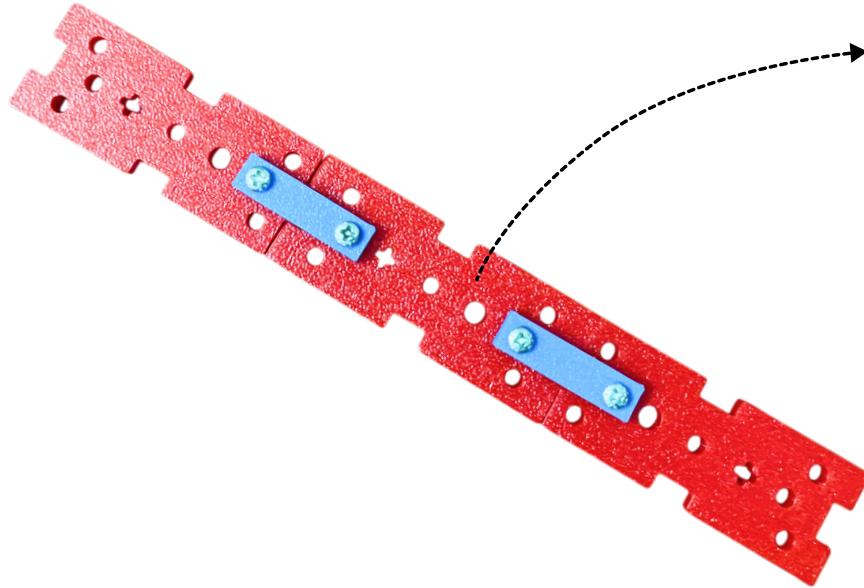


small screw - x8



nut - x8

step 1: assemble the body



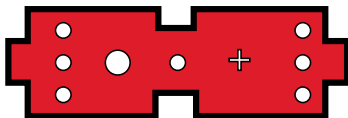
Lets join
2 x M-F beams
and a F-F beam.

Use 2 I-clamps to
hold them together.

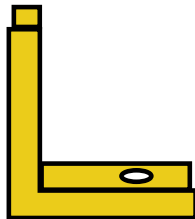


Repeat step 1 to make another set.

materials needed



m-m beam - x3



L clamp - x6

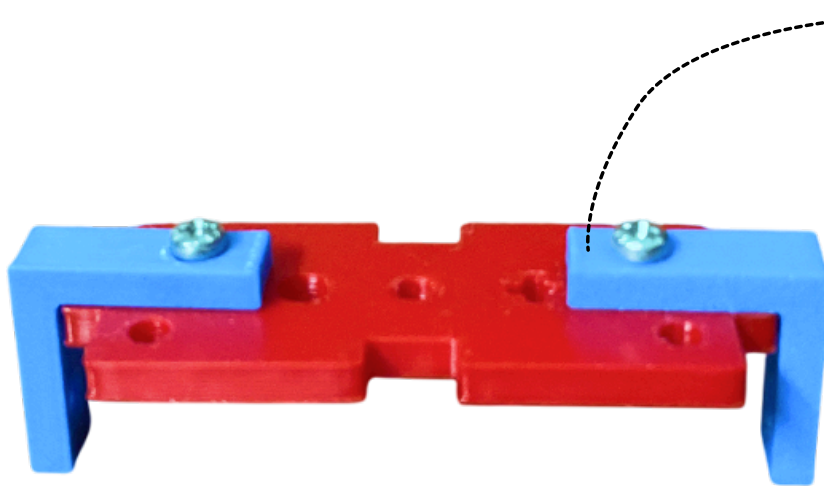


small screw - x6



nut - x6

step 2: assemble the supports

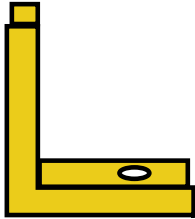


Attach **2x L clamps** to both ends of the **M-M beams**.



Repeat this step to make 2 more sets.

materials needed



L clamp - x1



small screw - x7



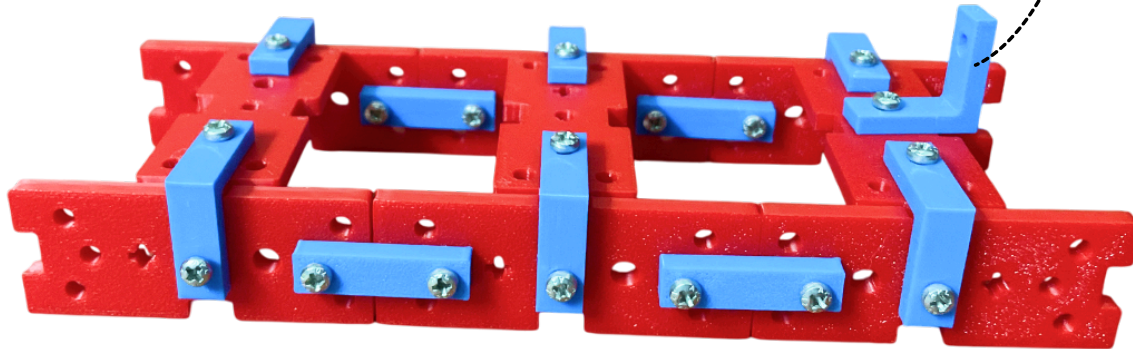
nut - x7

step 3: complete the body assembly

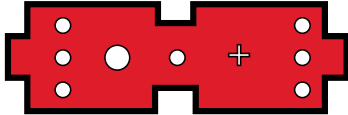
Join the pieces from
step 1 and 2 as shown.

With this step the
drawbridge's body
is complete.

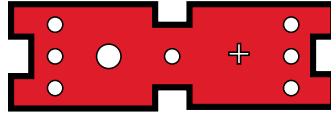
Attach a **L clamp**
to the support
as shown



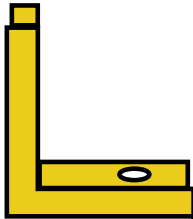
materials needed



m-m beam - x2



f-f beam - x2



L clamp - x5



small screw - x9



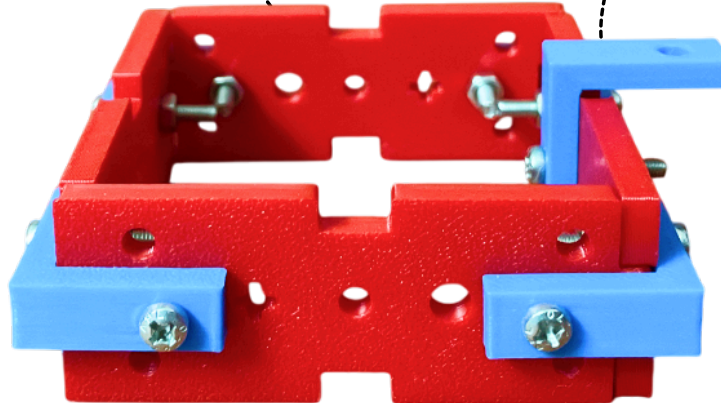
nut - x9

step 4: assemble the drawbridge's holder

Join **2 x F-F Beams** and **2 x M-M Beams** together to form a square.

Use **4 x L Clamps** to hold them together.

Fix a **L clamp** to the inside of the square as shown



materials needed



I clamp - x2

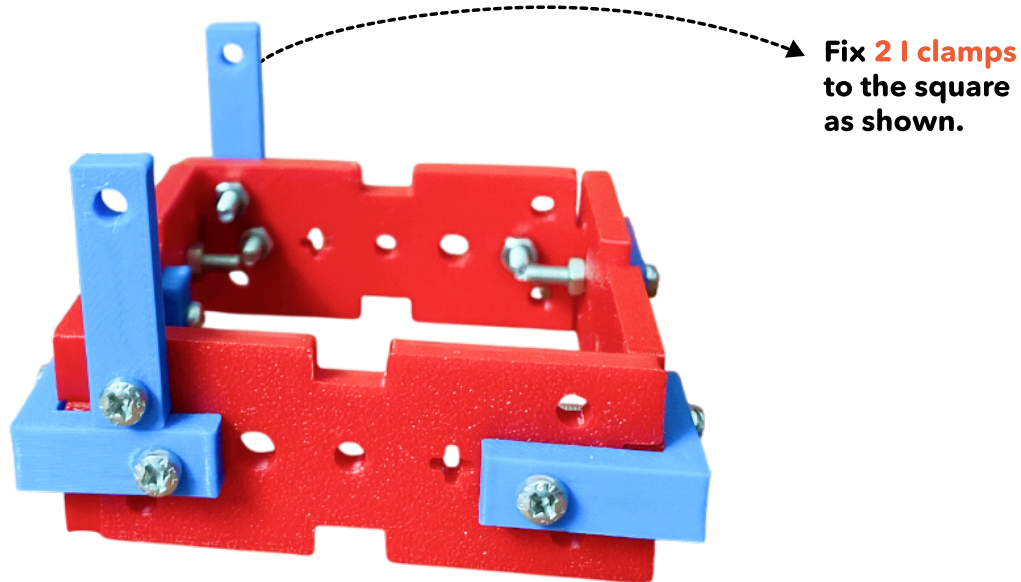


small screw - x2



nut - x2

step 5: assemble the axle holder

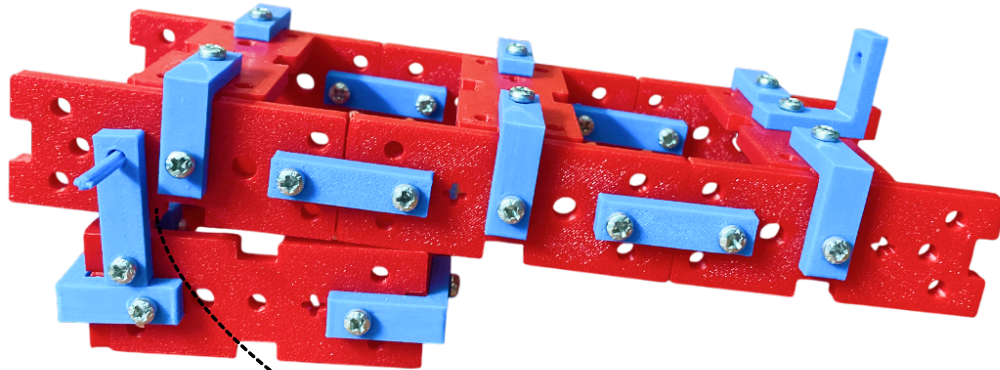


materials needed

long axle x1



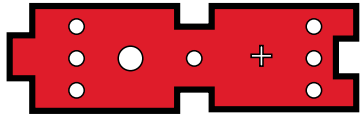
step 6: join the holder and bridge



→ Place the bridge in
between the holder.

Align the holes on the
I clamp and the bridge and
pass a **long axle** through it.

materials needed



m-f beam - x4



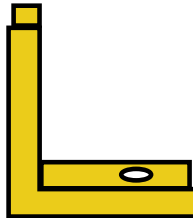
small screw - x7



nut - x7



I clamp - x2



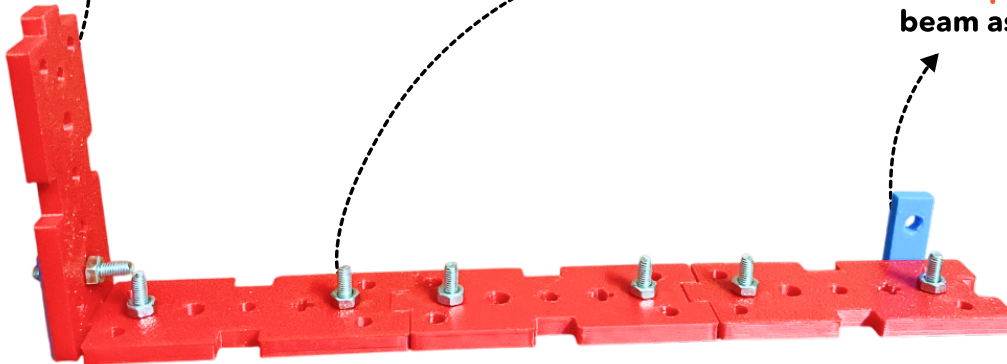
L clamp - x2

step 7: assemble the pulley system holder

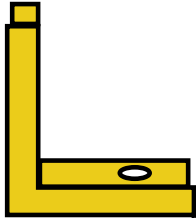
Attach a
M-F beam facing
upwards using
an **I clamp**

Join **3 x M-F beams**
together using
I clamps.

Then add another
L clamp to the other
beam as shown.



materials needed



L clamp - x1

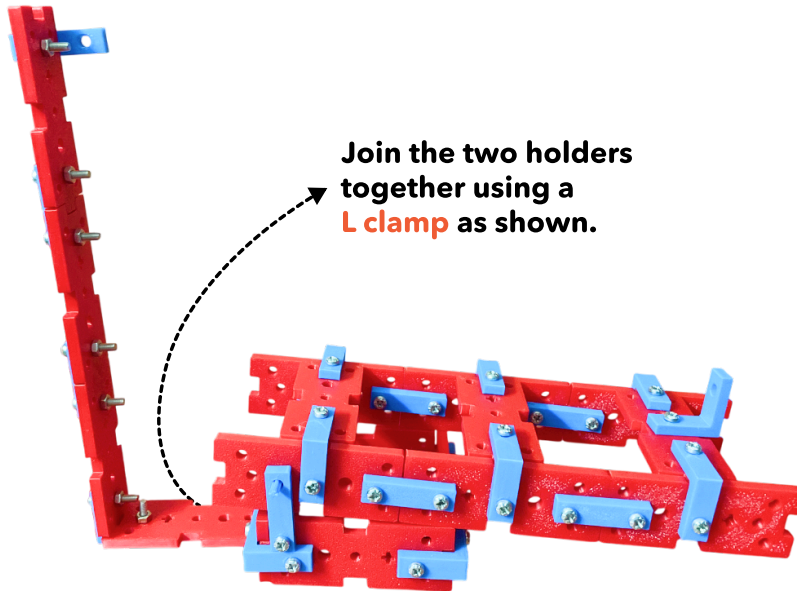


small screw - x2



nut - x2

step 8: attach the pulley holder & bridge holder



materials needed



gear box x1



long screw - x1

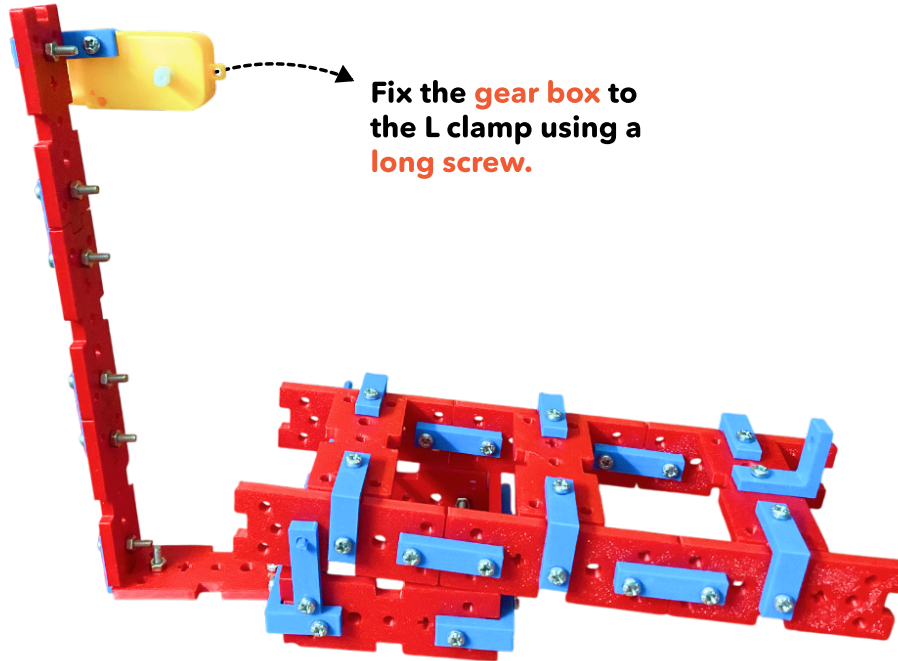


nut - x1



Do not tighten the gear box too much, keep it slightly loose so the gears inside can move easily.

step 9: add the gear box



materials needed



coupler - x2

step 10: complete the pulley system

